

Wangjie Su

Undergraduate | College of Engineering | Peking University

+86 188 6803 2449 | wsu0605@stu.pku.edu.cn

<https://github.com/csxq0605> | <https://csxq0605.github.io/>

Research Interests

Core question: How can foundation models become durable agents whose capabilities are definable, transferable, inspectable, and recoverable across models, Harnesses, and domains? I study capability/policy separation from execution adapters, long-horizon state and context, tool-use semantics, multi-agent coordination, failure recovery, and evaluation.

Applications and extensions: I use scientific simulation and experimental workflows as testbeds for agents that plan, execute, verify, and reproduce stateful processes. I aim to extend the same abstractions to embodied agents, where multimodal sensing, world models, and physical action form closed loops under partial observability and safety constraints.

Education

Peking University, B.S. in Theoretical and Applied Mechanics

Sep 2023 – Present

Peking University, Double Degree in Intelligent Science and Technology

Sep 2024 – Present

- **GPA:** 3.818/4.0
- **Selected Coursework:** Data Structures and Algorithms (100), Programming in AI (100), Computer Vision, Probability and Statistics, Numerical Analysis, Generative Modeling, Intelligent Robotics, Trustworthy ML, LLMs and NLG, and Physics Simulation for Computer Graphics.

Research Experience

AI + Fusion Energy Series Project

Undergraduate Researcher

Mar 2026 – Present

Xiniao Technology Innovation Center, Peking University | PIs: Prof. Yunfeng Liang and Prof. Songfang Huang

- **Problem:** Developing AI agents that can carry out long-horizon fusion-energy simulation workflows, from task planning and knowledge retrieval to tool execution, result interpretation, and iterative refinement.
- **Implementation:** Using **Nexgent** as the experimental harness to connect domain knowledge and simulation tools through MCP, coordinate specialist agents and explicit workflows, and preserve sessions, checkpoints, permissions, and tool traces.
- **Research focus:** Investigating long-context retrieval and recovery, sustained adherence and progress in long-horizon tasks, and permission-gated failure recovery so scientific runs remain state-consistent, auditable, and reproducible.

Combustion Laboratory, Peking University

Undergraduate Researcher

May 2025 – Present

PI: Prof. Zheng Chen

- **Background:** Studying fundamental combustion problems in energy conversion and propulsion, including flame dynamics, chemical kinetics, transport, ignition, flame propagation and stabilization, and laminar flame speed.
- **Current work:** Exploring mathematical formulations of multiscale combustion processes from conservation laws and reaction mechanisms, together with machine-learning approaches to predicting key quantities and approximating complex processes.
- **Methodological focus:** Comparing the regimes of mechanistic and data-driven models, with attention to physical consistency, interpretability, data efficiency, and out-of-distribution generalization toward physics–data integrated combustion modeling.

Internship Experience

Hangzhou Yuanxi Technology Co., Ltd.

AI Agent Developer Intern

Jul 2026 – Present

System: ManySelves — An Independently Designed General-Purpose Agent Capability Substrate

- **Capability definition:** Reframed an agent team as a versionable capability specification rather than a fixed application or prompt bundle: Identity contracts define roles, boundaries, I/O, and handoffs; Skill documents encode reusable methods and acceptance criteria; a stable runtime realizes both without domain-specific rewrites.

- **Execution abstraction:** Independently built the system end to end around one Main Agent and task-scoped specialists operating on project-local knowledge, state, and artifacts. Typed state, checkpoints, rollback, resumable decisions, and explicit review boundaries make long-horizon behavior inspectable and recoverable.
- **Research platform:** FastAPI exposes only conversation and task submission; model/tool orchestration, state transitions, and artifact provenance remain server-side. This creates a controlled substrate for studying capability transfer, multi-agent coordination, memory, and recovery in scientific and future embodied workflows.

Selected Projects

Nexgent: Scientific Agent Harness *Independent Project* **May 2026 – Present**

- **Problem:** Long-running scientific tasks need more than a chat interface: agents must connect to domain tools, coordinate specialized work, survive interruptions, expose risky actions, and leave a traceable experimental record. Nexgent was developed as the Agent Harness for prototyping these requirements in the AI + Fusion Energy project.
- **System:** Built a model-agnostic, frontend-neutral runtime shared by a PyQt6 desktop workspace, Textual TUI, and headless CLI; integrated 38 tools, automatic MCP tool bridging, multi-model routing, and sub-agent/workflow orchestration in pipeline, parallel, and phased modes.
- **Workflow functions:** Implemented progressive context compression, typed project/reference memory, session save–resume–fork, checkpoint rollback, permission gates, lifecycle hooks, goals, and background tasks for controlled and recoverable experiment execution.

oh-my-opencode (now oh-my-openagent) *Open-Source Contributor* **May 2026 – Present**

- **Research question:** Investigated which agent behaviors should remain invariant across OpenCode, Codex, Pi, and Claude Code rather than being entangled with one Harness. Proposed a downward-only Core–Adapter DAG separating reusable policy from Hook, provider, and UI effects.
- **Semantic abstractions:** Defined agent configuration as a zero-dependency core and model fallback as a session-scoped idle/armed/exhausted protocol with injected reachability and logging, turning implicit error handling into explicit, portable, and testable state transitions.
- **System evidence:** Preserved 40+ import sites through re-export shims and tested cleanup, re-arming, exhaustion, and session isolation. The maintainer confirmed that the work validated package extraction, which then moved into maintainer-led monorepo restructuring—a practical path toward Harness-independent policies for scientific and embodied backends.

Where Is My Codex (Flow Rescue) *Course Project* **Spring 2026**

- **Physics:** Co-developed a 2D fluid-puzzle game and implemented Position-Based Fluids in Python/Taichi with uniform-grid neighbor search, density constraints, terrain collisions, and GPU/CPU backends.
- **Engineering:** Built five playable levels and a Tiled import pipeline; added headless tests and profiling for digging, collision, channel stability, goal collection, grid construction, and constraint solving.

Tiny PyTorch *Course Project* **Fall 2024**

- **Framework:** Built a PyTorch-style tensor and reverse-topological autodiff engine with core operators, fully connected layers, convolution, pooling, softmax loss, and end-to-end MLP/CNN training.
- **Acceleration:** Implemented C++/CUDA storage, transfers, custom kernels, and cuBLAS-backed computation with Python bindings; trained on MNIST and compared single-GPU and DataParallel workflows.

Honors & Awards

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- National Scholarship and Merit Student, Peking University (2025)
 - Second Prize, Beijing Division, National College Student Mathematical Modeling Competition (2025)
 - Outstanding Learning Award and Third Scholarship, Peking University (2024)
 - First Prize, Beijing College Student Mathematics Competition (2024)

Skills

Programming: Python, C/C++, CUDA, MATLAB, LaTeX, CMake, Taichi

Tools & Frameworks: PyTorch, PyQt6, Git, Linux, Scikit-learn, Jupyter, OpenFOAM

Languages: Chinese (native); English (CET-6: 613)